

2018 SAJC Prelim Practical Exam Preparation List

Title	Q1 – Determination of percentage by mass of sodium carbonate in a mixture of sodium carbonate and sodium hydrogen carbonate		
Label Code	Chemicals	Preparation	Per Student
FA 1	0.0256 mol dm ⁻³ NaHCO ₃ + 0.0482 mol dm ⁻³ Na ₂ CO ₃	Dissolve 2.15 g NaHCO ₃ and 5.11 g Na ₂ CO ₃ in 1 dm ³	200 cm ³ (in capped plastic container)
FA 2	0.10 mol dm ⁻³ of hydrochloric acid, HCl		200 cm ³ (in capped plastic container)
Thymolphthalein indicator	Thymolphthalein	Dissolve 0.8 g of thymolphthalein powder in 500 ml of 96% ethanol and top up to 1 dm ³	1 bottle per candidate
Screened methyl orange indicator	Screened methyl orange		1 bottle per candidate
Apparatus (per student)	a) FA 1 and FA 2 in labelled capped plastic containers b) Filter funnel c) 25.0 cm³ pipette d) Burette x 1 e) Conical flask x 3 f) Dropping pipettes × 2 g) White tile h) Retort stand with clamp j) Distilled water bottle		

Title	Q2 Determination of n in the complex ion formula $[\text{Cu}(\text{NH}_3)_n]^{2+}$		
Label Code	Chemicals	Preparation	Per Student
FA 3	CuSO_4 solution	0.5 mol dm^{-3}	80 cm^3 (in capped plastic container)
FA 4	NH_3 solution	2.0 mol dm^{-3}	80 cm^3 (in capped plastic container)
Apparatus (per student)	a) 2 boiling tubes b) 2 50.00 cm^3 burettes c) 1 thermometer d) 1 glass funnel e) Paper towels		

Title	Q3 identifying cations in an unknown sample		
Label Code	Chemicals	Preparation	Per Student
FA 3	CuSO ₄ + Fe ₂ (SO ₄) ₃ solution	1.5 mol dm ⁻³ CuSO ₄ 0.5 mol dm ⁻³ Fe ₂ (SO ₄) ₃ Filter to remove the undissolved CuSO ₄ .	30 cm ³ (in capped plastic container)
FA 1	Mixture of Na ₂ CO ₃ and NaHCO ₃	See Q1	See Q1
Potassium iodide	Solid KI		5 g
Hexane	Hexane		10 cm ³
Aqueous ammonia	NH ₃ (aq)	Std. preparation Ammonia needs to be made fresh.	Bench reagent
Aqueous sodium hydroxide	NaOH (aq)	Std. preparation	Bench reagent
Sulfuric acid	H ₂ SO ₄ (aq)	Std. preparation	Bench reagent
Nitric acid	HNO ₃ (aq)	Std. preparation	Bench reagent
Hydrochloric acid	HCl (aq)	Std. preparation	Bench reagent
Barium nitrate	Ba(NO ₃) ₂ (aq)	Std. preparation	Bench reagent
Silver nitrate	AgNO ₃ (aq)	Std. preparation	Bench reagent
Potassium manganate(VII)	KMnO ₄ (aq)	Std. preparation	Bench reagent

Limewater Ca(OH)₂	Ca(OH)₂	Std. preparation	Bench reagent
Apparatus (per student)	a) 1 × glass rod; b) 1 × white tile; c) 1 × glass funnel; d) 4 × dropping pipettes; e) 1 × wash bottle containing deionised water; f) 6 x test tubes g) 2 x boiling tubes h) 1 x test tube rack i) 1 x test tube brush j) 1 x tongs k) 1 x Bunsen burner l) 1 x delivery tube m) 2 x wooden splints n) 1 x goggles o) 1 x marker pen p) Filter paper q) Filter Paper/ Litmus Box r) 1 x Lighter s) Paper towels		